Problem 8.6: Uniform flow depth is 3m, velocity is 2m/s and assume the channel is wide and rectangular.

a) Determine the flow regime.

b) A sill constructed on the bed has 25cm height. Show, is it possible the flow past over the sill? What is the water depth on the sill?

c) What is maximum (critical) height of the sill?

Solution:

a) For rectangular channel (q=

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{36}{9.81}} = 1.54 < 3 \text{ m}$$ flow is in sub-critical regime.

$q = 3 \times 2 = 6 \text{ m}^3/\text{sm}$

or

$$Fr = \frac{V}{\sqrt{gy}} = \frac{2}{\sqrt{9.81 \times 3}} = \frac{2}{5.42} = 0.37 < 1$$

b)

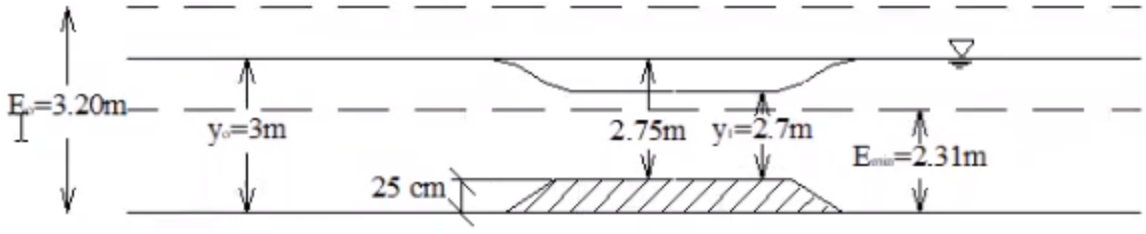


Figure p8.6

$$E_{min} = \frac{q^2}{2gy_c^2} + y_c = \frac{gy_c^3}{2gy_c^2} + y_c = \frac{y_c}{2} + y_c = \frac{3}{2}y_c$$

$$E_{min} = \frac{3}{2}y_c = \frac{3}{2} \times 1.54 = 2.31\text{m}$$

$$E_0 = \frac{q^2}{2gy^2} + y = \frac{36}{19.62 \times 9} + 3 = 3.20 \text{ m}$$

$$E_0 = E_1 + e$$

$$E_1 = E_0 - e = 3.20 - 0.25 = 2.95 \text{ m}$$

$E_1 > E_{min}$ Water depth will not increase at upstream of the sill

Water depth on the sill y_1 ;

$$2.95 = y_1 + \frac{q^2}{2gy_1^2} = y_1 + \frac{36}{19.62y_1^2} = y_1 + \frac{1.8}{y_1^2}$$ by trial and error;

For y_1=2 m $\quad 2 + \frac{1.8}{4} = 2.45 \neq 2.95$

For y_1=2.5 m $\quad 2.5 + \frac{1.8}{6.25} = 2.79 \neq 2.95$

For y_1=2.6 m $\quad 2.6 + \frac{1.8}{6.76} = 2.87 \neq 2.95$

For y_1=2.7 m $\quad 2.7 + \frac{1.8}{7.29} = 2.95 = 2.95$

So y_1=2.7 m.

c) E_{min} does not depend on height of sill but the energy level related with y. $e=e_{max}$ for $E_1=E_{min}$

$E_0 \geq E_{min}+e_{max} \rightarrow e_{max} = E_0 - E_{min} = 3.20 - 2.31 = 0.89$ m $\leq$ 89 cm

Problem 8.7: Determine flow depth and energy level at upstream of the sill of 1m height constructed on channel bed for previous problem. Draw the water surface profile.

Solution:

a) $E_0 = E_{min} + e = 2.31 + 1 = 3.31$ m

$$E_0 = y_0 + \frac{q^2}{2gy_0^2} \qquad 3.31 = y_0 + \frac{1.8}{y_0^2} \qquad \text{by trial and error}$$

For y_0=3.2 m $\qquad 3.2 + \frac{1.8}{10.24} = 3.375 \neq 3.31$

For y_0=3.15 m $\qquad 3.15 + \frac{1.8}{9.92} = 3.33 \neq 3.31$

For y_0=3.14 m $\qquad 3.14 + \frac{1.8}{9.86} = 3.32 \neq 3.31$

For y_0=3.13 m $\qquad 3.13 + \frac{1.8}{9.7969} = 3.31 = 3.31$

y_0=3.13 m

so, the depth is 13 cm at upstream of the sill.

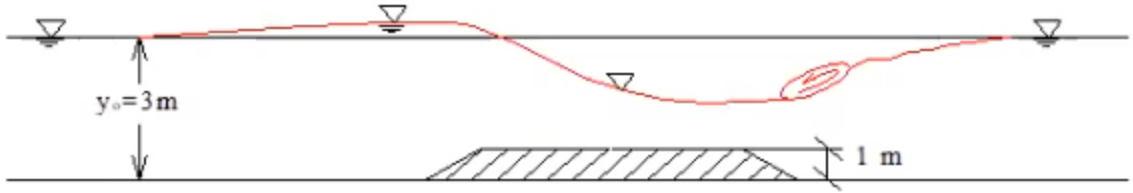


Figure p8.7

Problem 8.8: Uniform flow depth and velocity are y_0=1 m and V=5 m/s respectively;

a) Determine the flow regime

b) If a sill constructed on the channel bed has 10 cm height, does discharge flow over the sill with existing energy? Determine the water depth over the sill?

c) What is critical height of the sill?

d) If height of the sill constructed on the channel bed is 40 cm, calculate the water depth at upstream section of the sill.

Solution:

a) $y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{25}{9.81}} = 1.37 > 1\text{m}$ super critical flow

or

$$Fr = \frac{V}{\sqrt{gy}} = \frac{5}{\sqrt{9.8 \times 1}} = 1.60 > 1$$

b) $E_0 = y_0 + \frac{q^2}{2gy_0^2} = 1 + \frac{25}{19.62 \times 1^2} = 2.25\,\text{m}$

$$E_{min} = \frac{3}{2} y_c = \frac{3}{2} \times 1.37 = 2.06\text{m}$$

$$E_1 = E_0 - e = 2.25 - 0.10 = 2.15\,\text{m}$$

$E_1 > E_{min}$ discharge flows with existing energy.

The water depth over the sill;

$$2.15 = y_1 + \frac{q^2}{2gy_1^2} = y_1 + \frac{1.25^2}{y_1^2}$$

By trial and error y_1 may be found. It must be less than 1.37m because the flow regime is supercritical.

For y_1=1.2 m $\quad 1.2+\frac{1.25}{1.44}=2.07\neq 2.15$

For y_1=1.3 m $\quad 1.3+\frac{1.25}{1.69}=2.07\neq 2.15$

For y_1=1.1 m $\quad 1.1+\frac{1.25}{1.21}=2.13\neq 2.15$

For y_1=1.09 m $\quad 1.09+\frac{1.25}{1.19}=2.15=2.15$

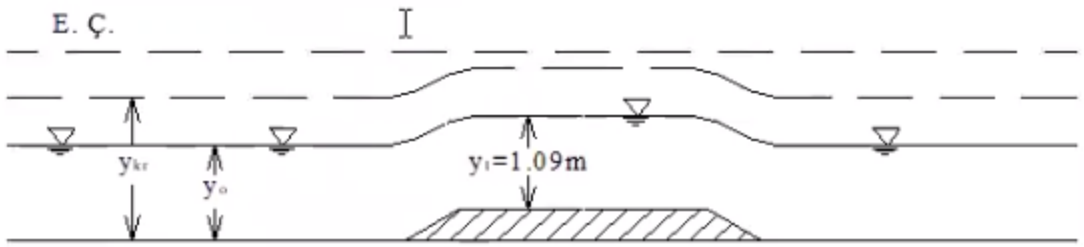


Figure p8.8a

c) $E_0=E_{min}+e_{max}$

$e_{max}=E_0-E_{min}=2.25-2.06=0.19$ m= 19 cm

d) $E_0=E_{min}+e=2.06+0.40=2.46$ m

$$2.46=y_0^*+\frac{q^2}{2gy_0^{*2}}=y_0^*+\frac{1.25}{y_0^{*2}}$$

By trial and error y_0^* may be found and assume $y_0^*>y_{kr}$.

c) $E_0 = E_{min} + e_{max}$

$e_{max} = E_0 - E_{min} = 2.25 - 2.06 = 0.19\ m = 19\ cm$

d) $E_0' = E_{min} + e = 2.06 + 0.40 = 2.46\ m$

$$2.46 = y_0^* + \frac{q^2}{2gy_0^{*2}} = y_0^* + \frac{1.25}{y_0^{*2}}$$

By trial and error y_0^* may be found and assume $y_0^* > y_{kr}$. $y_0^* = 2.16m$

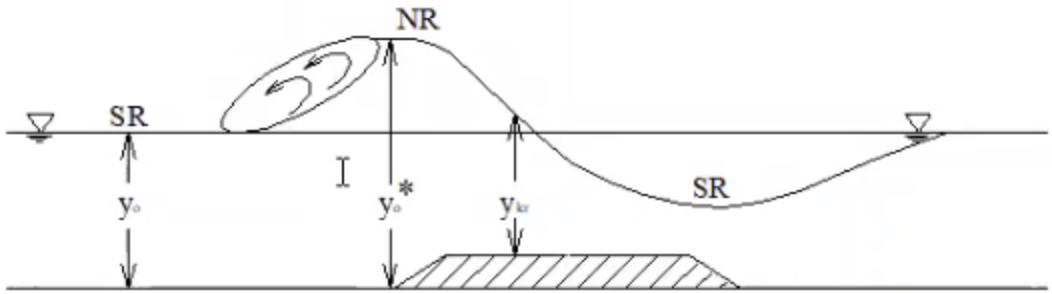


Figure p8.8b

Problem 8.9: Discharge flows in a channel with 10m width. Depth is 3 m and flow velocity is 2 m/s.

a) Determine the flow regime

b) Determine the minimum channel width with existing energy?

c) Draw the water profile.

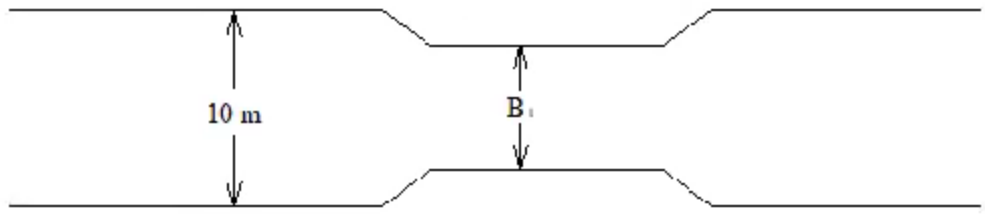


Figure 8.9

Solution:

a) $Q - 2 \times 3 \times 10 - 60\ m^3/s \rightarrow q - 60/10 - 6\ m^3/s$

$y_c = \sqrt[3]{\frac{36}{9.81}} = 1.54 < 3m$ Sub-critical regime

b) $E_0 = y_0 + \frac{q^2}{2gy_0^2} = 3 + \frac{36}{19.62 \times 9} = 3.2\,\text{m}$ q=Vxy

$$E_{min} = \frac{3}{2} y_c$$

$$3.2 = \frac{3}{2} y_{kr} = \frac{3}{2} \sqrt[3]{\frac{Q^2}{B_1^2 g}} = \frac{3}{2} \sqrt[3]{\frac{3600}{B_1^2 \times 9.81}} \rightarrow B_1 = 6.10\,\text{m}$$

$$q_1 = \frac{60}{6.10} = 9.83\ \text{m}^3/\text{sm}$$

$$3.20 = y_1 + \frac{9.83^2}{19.62 y_1^2}$$

$$2.20 = y_1 + \frac{4.92}{y_1^2}$$ by trial and error y_1=2 m

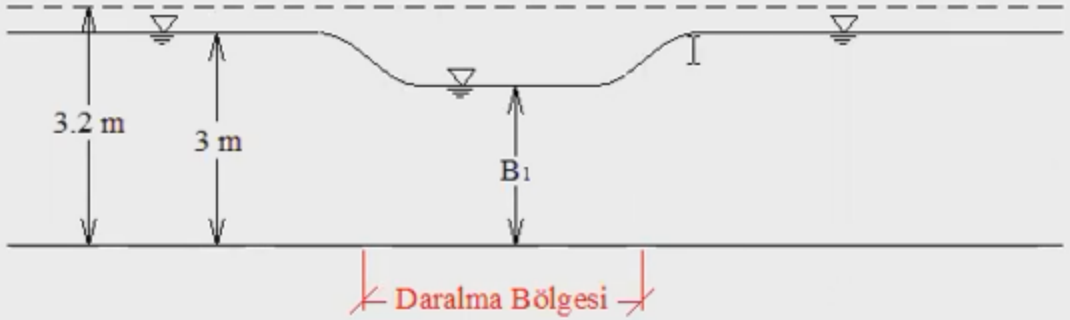
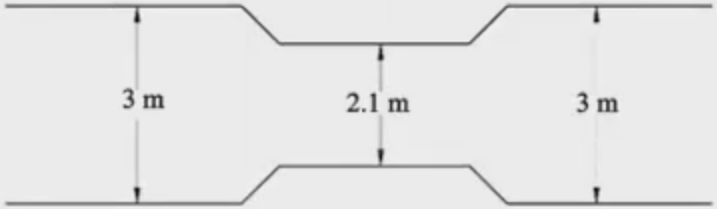
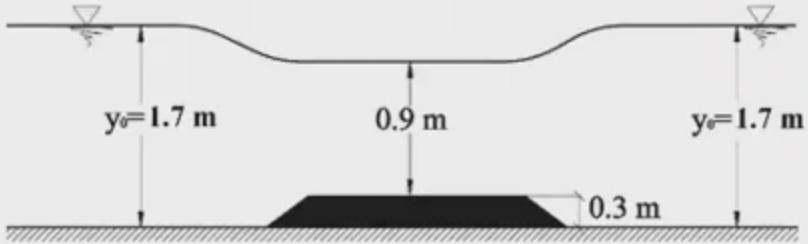


Figure p8.9b

Problem 8.11: 7.2 m^3/s flowrate flows in a rectangular channel shown in the Figure below. Is water surface level shown in the cross-section possible?. Calculate the water level.



Plan



Cross-section

$$q_0 = \frac{7.2}{3} = 2.4m^2/s \rightarrow E_0 = 1.7 + \frac{2.4^2}{19.62 \times 1.7^2} \rightarrow E_0 = 1.80m$$

$$q_1 = \frac{7.2}{2.1} = 3.43m^2/s \rightarrow E_1 = 0.9 + \frac{3.43^2}{19.62 \times 0.9^2} \rightarrow E_1 = 1.64m$$

$1.80 \neq [1.64 + 0.3]$ $1.94\ E_0 = E_1 + e$ olmalılmaıdeya O kesitinde $Fr_0 = \dfrac{\dfrac{q_0}{y_0}}{\sqrt{gy_0}} = \dfrac{\dfrac{2.4}{1.7}}{\sqrt{9.81 \times 1.7}}$

Sekilde görülen su yüzü profilinin oluşması için $E_0 = E_1 + e$ olmalıdır veya

$$\text{O kesitinde } Fr_0 = \frac{\frac{q_0}{y_0}}{\sqrt{gy_0}} = \frac{\frac{2.4}{1.7}}{\sqrt{9.81\times 1.7}} \Rightarrow Fr_0 = 0.35 < 1 \quad \text{NR}$$

$$\text{1 kesitinde } Fr_1 = \frac{\frac{q_1}{y_1}}{\sqrt{gy_1}} = \frac{\frac{3.43}{0.9}}{\sqrt{9.81\times 0.9}} \Rightarrow Fr_0 = 1.28 > 1 \quad \text{SR}$$

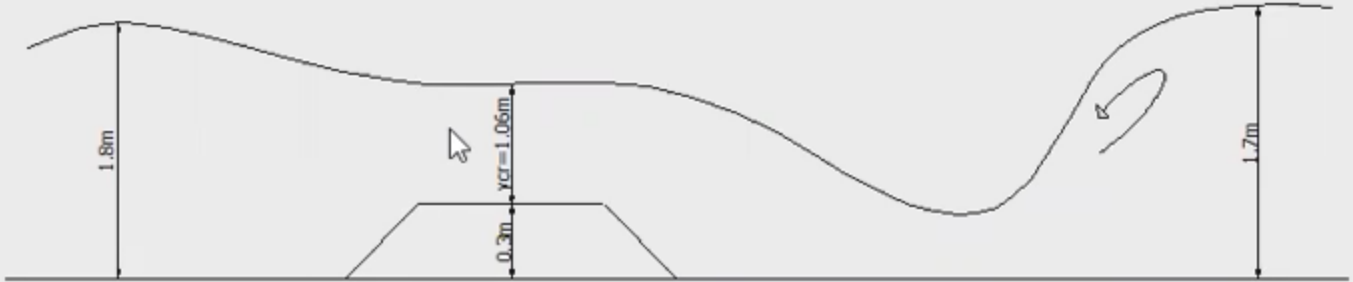
Dolayısı ile sekilde verilen su yüzünün oluşması mümkün değildir.

$$y_{cr1} = \sqrt[3]{\frac{3.43^2}{9.81}} = 1.06m \rightarrow E_{\min} = \frac{3}{2} y_{cr1} = \frac{3}{2}\times 1.06 \rightarrow E_{\min} = 1.59m$$

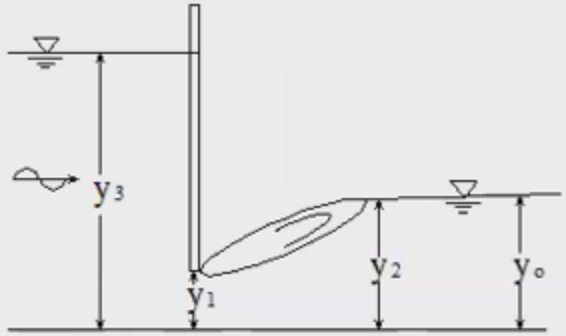
$$E_0' = 1.59 + 0.3 = 1.89m$$

$$E_0' = y_0' + \frac{q^2}{2g\times y_0'^2} \rightarrow 1.89 = y_0' + \frac{2.4^2}{19.62\times y_0'^2} \rightarrow y_0' = 1.80m$$

Görüldüğü gibi esik üzerinde kritik akım kosulları oluşmaktadır. Bu durumda su yüzü aşağıdaki sekilde görüldüğü gibi olacaktır.


1.8m
ycr=1.06m
0.3m
1.7m

Problem 8.12: A vertical sluice gate constructed in a rectangular channel with 6 m width, 0.0004 bottom slope and n=0.015. If discharge is 8 m^3/s and opening cause a hydraulic jump, calculate jump depths and force on the gate (neglect energy loss).



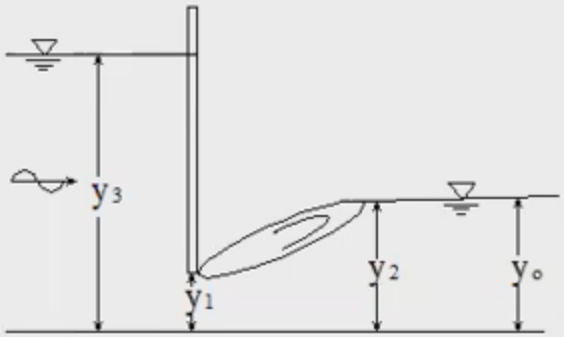


Figure p8.12

$$q = \frac{8}{6} = 1.33 \text{ m}^3/\text{msn}$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{1.33^2}{9.81}} = 0.56 \text{ m}$$

$$8 = \frac{1}{0.015} \times \left(\frac{6 \times 0.56}{6 + 2 \times 0.56} \right)^{2/3} \times J_c^{1/2} \times 6 \times 0.56$$

$J_c = 0.0035$ ve $J_0 > J_c$ SR

Uniform flow depth;

$$Q = \frac{1}{n} R^{2/3} J_0^{1/2} A \qquad 8 = \frac{1}{0.015} \times \left(\frac{6y_0}{6 + 2y_0} \right)^{2/3} \times 0.02 \times 6y_0$$

$$8 = 8 \times \left(\frac{6y_0}{6 + 2y_0} \right)^{2/3} \times y_0$$

$$1 = \left(\frac{6y_0 y_0^{3/2}}{6 + 2y_0} \right)^{2/3}$$

$$1 = \left(\frac{6y_0^{5/2}}{6 + 2y_0} \right)^{2/3}$$ by trial and error $y_0 = 1.14$ m, $y_2 \cong y_0 = 1.14$ m .

$$\frac{y_1}{1.14} = \frac{1}{2}\left(-1 + \sqrt{1 + 8 \times 0.35^2}\right) = 0.20$$

$$V_2 = \frac{Q}{A} = \frac{8}{6 \times 1.14} = 1.17 \text{ m/s}$$



$$Fr_2 = \frac{V_2}{\sqrt{gy_2}} = \frac{1.17}{\sqrt{9.81 \times 1.14}} = 0.35$$

y_1=0.23 m

Force on the gate

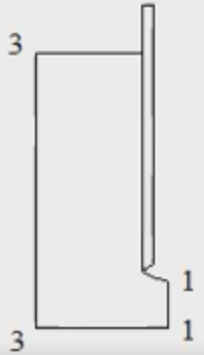


Figure p8.10b

Figure p8.106



$$V_1 = \frac{Q}{b_1 y_1} = \frac{8}{6 \times 0.23} = 5.8 \text{ m/s}$$

$$y_3 = \frac{V_1^2}{2g} + y_1 = \frac{5.8^2}{19.62} + 0.23 = 1.94 \text{m}$$

$$\rho g \frac{y_3^2}{2} - \rho g \frac{y_1^2}{2} - \rho Q V_1 - F = 0$$

$$F = 10^4 \times \frac{1.94^2}{2} - 10^4 \times \frac{0.23^2}{2} - 10^3 \times 8 \times 5.8$$

$$F = -28199 \text{ N}$$

$$\bar{F} = 28.2 \text{ KN}$$